Weekly Progress Report: SLM-to-SLM Guided Reasoning

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***Abstract*—This report covers six contributions from this week. The guided pipeline was replicated with Llama 3.2, confirming generalisation across model families. The GSM8K-trained Llama guide transfers to diverse datasets, with ASDiv planned as the next test. FT Qwen2.5-3B and Phi-3.5-mini were validated as solo baselines, a data quality ablation identified verbal richness as the key factor, and initial publication venues were surveyed.**

***Index Terms*—small language models, LoRA fine-tuning, guided reasoning, Llama, cross-architecture, GSM8K**

I. Project Overview

This research investigates whether a fine-tuned SLM can replace expensive LLM guides in a guided reasoning pipeline. A LoRA fine-tuned [3] guide produces verbal plans consumed by a 5-vote solver ensemble [2]. Prior weeks confirmed gains on five datasets using Qwen2.5. This week extends experiments to Llama 3.2 and closes remaining controls.

II. Completed Work This Week

A. Llama 3.2 Pipeline Generalisation

The pipeline was run with Llama 3.2-3B (fine-tuned guide, LoRA on GSM8K) and Llama 3.2-1B (solver, 5 votes) at 8.0B param-passes, 46.7% below the natural ceiling (Table I).

TABLE I
Llama 3.2 Pipeline Results vs. Qwen Reference.

Dataset	N	Base	Guided	Llama Δ	Qwen Δ
SVAMP	300	28.7%	49.7%	+21.0	+21.3
ARC-C	900	47.6%	58.1%	+10.6	+10.6
ASDiv	300	39.0%	50.3%	+11.3	+10.3
CSQA	900	49.0%	54.4%	+5.4	+5.1
RACE-High	700	41.0%	46.7%	+5.7	−0.2
PIQA	900	61.9%	56.4%	−5.4	+24.2

Gains on arithmetic and commonsense tasks closely match Qwen. The PIQA result inverts from $+24.2$ pp (Qwen) to -5.4 pp (Llama, p=0.031), traced to a position bias in the Qwen solver (93.4% A-selection) that the guide happened to correct. The Llama solver has no such bias, so the guide adds noise instead. The PIQA gain is reclassified as a **position-bias artefact**.

B. GSM8K-Trained Guide: Generalisation Scope

The Llama guide is trained only on GSM8K [4] yet generalises to ARC-C ($+10.6$ pp), CSQA ($+5.4$ pp), and RACE-High ($+5.7$ pp). The transferable property is the structural reasoning scaffold of GSM8K's multi-sentence problems, not arithmetic knowledge. ASDiv testing is planned but not yet done.

C. FT Phi-3.5-mini-instruct: Cross-Architecture Validation

The same LoRA rank-8 protocol on Phi-3.5-mini (3.82B, only 0.08% trainable) achieves 75.3% average, exceeding both FT Qwen-3B (73.8%) and the guided pipeline (72.0%) at the same compute reduction. Fine-tuning gains are **architecture-agnostic**.

D. Fine-Tuning Data Quality Ablation

A guide trained on ASDiv (short, template-like problems) underperforms the GSM8K-trained guide by 46% on average gain, even on its own test set. Domain match is a weak predictor; **verbal richness and structural diversity** of the training data are what matter.

E. Publication Venue Survey

We explored potential venues where this work could be submitted. Some initial ideas have been gathered on suitable publication targets.

III. Remaining Tasks

- **ASDiv Generalisation Test:** Evaluate the Llama 3.2 pipeline on ASDiv.
- **Error Analysis:** Categorise guided-wrong/baseline-correct cases to distinguish genuine planning benefit from bias-correction effects.

References

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